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SCHOOL OF COMPUTER SCIENCE & ENGINEERING

Question Bank

Subject Name: - Mathematical Concepts For AI

Class: B.Tech AIML, 1st Year

Semester: 1st Sem

Multiple-Choice Questions (MCQs)

Q1. If $2x/5 = 4$, the value of x is-

- a. 10
- b. -10
- c. -8/5
- d. 8/5

Q2. What is the slope of the straight line $y = -3x + 2$?

- a. 2
- b. 3
- c. -3
- d. -2

Q3. Two lines are said to be parallel if the difference in their slope is

- a. 1
- b. 0
- c. -1
- d. None of the above

Q4. Two lines are said to be perpendicular if the product of their slope is equal to

- a. -1
- b. 1
- c. 0
- d. 12

Q5. The equation of a line that passes through the points (1, 5) and (2, 3) is

- a. $2x - y - 7 = 0$
- b. $2x + y + 7 = 0$
- c. $2x + y - 7 = 0$
- d. $x + 2y - 7 = 0$

Q6. If a line has slope 3 and pass through point (1, 2) then the equation of line is

- a) $x=3y-1$
- b) $x=3y+1$
- c) $y=3x+1$
- d) $y=3x-1$

Q7. Log X = 1000 then find the value of X

- a) 1
- b) 2
- c) 3
- d) None of these.

Q8. Find the derivative of $3X^2+4X-12$

- a) $6X+8$
- b) $-6X+4-12$
- c) $6X+4-12$
- d) $6X+4$

Q9. Find the $\lim_{x \rightarrow 2} \left(\frac{3x-6}{x-2} \right)$

- a) Not defined
- b) 0
- c) Infinite
- d) 3**

Q10. Solve $\lim_{x \rightarrow 3} \left(\sin \frac{3x}{x} \right)$

- a. 4
- b. 3**
- c. 1
- d. 2

Q11. Determine the derivative of the function: $x^2 \sin x$

- a. $2x \sin x - x^2 \sin x$
- b. $2x \cos x - x^2 \sin x$
- c. $2x \sin x + x^2 \cos x$**
- d. $\cos x - x^2 \sin x \cos x$

Q12. The side of an equilateral triangle is increasing at the rate of 3 cm/s. The rate at which area increases when the side is 30 is

- a. $45\sqrt{3} \text{ cm}^2/\text{s}$**
- b. $15/3 \text{ cm}^2/\text{s}$
- c. $\sqrt{3} \text{ cm}^2/\text{s}$
- d. $5\sqrt{3} \text{ cm}^2/\text{s}$

Q13 A polynomial of degree n has

- a. No zero
- b. n zeros**
- c. no solution
- d. n solutions

Q14. One of the solutions of $|x| < 1$ is

- a. -1
- b. 1
- c. 0**
- d. 2

Q15. Mean of 12,18,20,30,20

- a. 25
- b. 20**
- c. 30
- d. 28

Q16. Median of the given data 10,20,30,40,50,25,35

- a. 20
- b. 25
- c. 30**
- d. 40

Q17. Solution of the given equation $2X - 4 = 6$

- a. 2
- b. 4
- c. 5**
- d. 6

Q18. If $f(X) = 3X^2 - 2X + 4$ Find $f(2)$

- a. 4
- b. 10
- c. 5
- d. 12**

Q19. If $f(X) = 3X^3 - 3X + 4$ Find $f(2)$

- a. 4
- b. 0
- c. 22**
- d. 24

Q20. Solution of the given equation $3X + 5 = 11$

- a. 2**
- b. 5
- c. 3
- d. 4

Q21. $\log 1000 = X$ Find the value of X .

- a. 2
- b. 3**
- c. 4
- d. 5

Q22. $\log X = 2$ Find the value of X .

- a. 10
- b. 1000
- c. 100**
- d. 0

Q23. If the mean of 4, 6 and X is 6 find the value of X

- a. 4
- b. 2
- c. 14
- d. 8**

Q24. If the mean of 8, 10 and X is 12 find the value of X

- a. 10
- b. 18**
- c. 12
- d. 20

Q25. What is a vector?

- a) A scalar quantity
- b) A quantity with magnitude and direction**
- c) A two-dimensional matrix
- d) None of the above

Q26. What is the probability of an impossible event?

- a) 0**
- b) 0.5
- c) 1
- d) Cannot be determined

Q27. What is the median of the data set: 10, 15, 20, 25, 30?

- A) 10
- B) 20**
- C) 25
- D) 15

Q28. If you flip a fair coin, what is the probability of getting heads?

- A) 0.25
- B) 0.5**
- C) 1
- D) 0.75

Q29. If probability of happening is 0.56 then what is the probability of not happening?

- a) 0.5
- b) 1
- c) 0
- d) 0.44**

Q30. Find the mean of 6, 12, 24, 8 and 15

- a) 12
- b) 13**
- c) 15
- d) 10

Short Question Answers

Unit – 1

Q1. Calculate the slope of a line, that passes through the origin, and the mid-point of the segment joining the points P (0, -4) and B (8, 0).

Q2. Solve the following system of equations

$$2X + 3Y = 7$$

$$4X + Y = 9$$

Q3. Find the equation of a straight line which is passing through points (2, -5) and (-4, 7)

Unit – 2

Q1. Suppose

$$f(x) = \begin{cases} a + bx, & x < 1 \\ 4, & x = 1 \\ b - ax, & x > 1 \end{cases} \quad \text{and if } \lim_{x \rightarrow 1} f(x) = f(1)$$

What are the possible values of a and b?

Q2. Find the derivative of the following function from the first principle.

$$F(x) = (X-1)(X-2)$$

Q3. For the given curve: $y = 5x - 2x^3$, when x increases at the rate of 2 units/sec, then how fast is the slope of curve changes when $x = 3$?

Q4. Find the value of x for which the points (x, -1), (2, 1) and (4, 5) are collinear.

Q5. Find the equation of the line passing through (-3, 5) and perpendicular to the line through the points (2, 5) and (-3, 6).

Q6. Find the equation of the line passing through (1, 2) and (3, -4).

Unit – 3

Q10. Find the eigen values for the given matrix $\begin{bmatrix} 2 & 1 \\ 4 & 5 \end{bmatrix}$

Q11. Find the eigen values for the given matrix $\begin{bmatrix} 2 & 0 \\ -1 & 1 \end{bmatrix}$

Q12. Multiply the given matrices

$$A = \begin{bmatrix} -3 & 4 \\ 5 & 1 \\ 2 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 5 & -2 & 2 \\ 4 & 0 & -3 \end{bmatrix}$$

Q13. Solve the following system of equations-

$$2x + 3y = 12, 3x - 2y = 5$$

Unit – 4

Q14. Find the dot and cross products of two vectors ($\vec{A}$ and $\vec{B}$)

$$\vec{A} = -2\mathbf{i} + 3\mathbf{j} + 4\mathbf{k}$$

$$\vec{B} = 3\mathbf{i} - 4\mathbf{j} - 2\mathbf{k}$$

Q15. A card has been drawn from a full pack of playing card. Find the probability of getting a black king?

Q16. Define the following terms-

- Population
- Sampling
- Data
- Statistics in singular sense

Q17. Find the probability of given conditions

- Getting a number greater than 3
- Getting an even number.
- Getting a number less than 4.

Unit – 5

Q18. Find Mean, Median and IQR from the given data

40,60,50,30,20,30,70

Q19. Find Mean, Median and IQR from the given data

45,55,70,65,25,30,85

Long Question Answers

Unit – 1

Q1. Solve the following system of equations

$$\begin{array}{ll} \text{a) } 3X + 4Y = 5 & \text{b) } 3X + 5Y = 7 \\ 4X + 6Y = 9 & 2X - 2Y = 1 \end{array}$$

Q2. The perpendicular from the origin to the line $y = mx + c$ meets it at the point $(-1, 2)$. Find the values of m and c .

Q3. Find the equation of a straight line which is perpendicular to the $2Y = 3X - 4$ and passing through $(-2, 5)$

Unit – 2

Q1. The side of an equilateral triangle is increasing at the rate of 2 cm/s. The rate at which area Increases when the side is 20cm.

Q2. The volume of a sphere is increasing at the rate of 8 cm³/s. Find the rate at which its surface area is increasing when the radius of the sphere is 12 cm.

Unit – 3

Q1. Solve the system of linear equations by gauss elimination method Consider the following system of linear equations in three variables.

$$2x - y + 3z = 9$$

$$x - 3y - 2z = 0$$

$$3x + 2y - z = -1$$

Q2. Solve the system of linear equations by gauss elimination method Consider the following system of linear equations in three variables.

$$X + Y + Z = 3$$

$$X + 2Y + 3Z = 0$$

$$X + 3Y + 2Z = 3$$

Unit – 4

Q1. Suppose you roll a fair six-sided die.

- (a) What is the sample space for this experiment?
- (b) What is the probability of rolling a number greater than 4?
- (c) What is the probability of rolling an odd number?

Q2. A bag contains 3 red balls and 2 green balls. A ball is drawn at random.

- (a) What is the probability of drawing a red ball?
- (b) If the ball is not replaced, what is the probability of drawing a green ball on the second draw?
- (c) If the ball is replaced, what is the probability of drawing a red ball on the second draw?

Q3. What is the probability of taking one card out from full pack of playing card.

- a) Getting red card.
- b) Getting red face card.
- c) Getting red queen.
- d) Getting black card.
- e) Getting an ace.

Unit – 5

Q1. Find Mean, Median and IQR from the given data

45,60,80,65,25,30,85,20,25,35,40

Q2. Find Mean, Median and IQR from the given data

50,65,60,40,25,35,75,30,90,45,35.

Question Bank AI&ML

Q1. If $2x/5 = 4$, the value of x is-

- I. 10
- II. -10
- III. $-8/5$
- IV. $8/5$

Q2. What is the slope of the straight line $y = -3x + 2$?

- a. 2
- b. 3
- c. -3
- d. -2

Q3. Two lines are said to be parallel if the difference in their slope is

- a. 1
- b. 0
- c. -1
- d. None of the above

Q4. Two lines are said to be perpendicular if the product of their slope is equal to

- a. -1
- b. 1
- c. 0
- d. 12

Q5. The equation of a line that passes through the points (1, 5) and (2, 3) is

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- c) $y=3x+1$
- d) $y=3x-1$

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Q12. The side of an equilateral triangle is increasing at the rate of 3 cm/s. The rate at which area increases when the side is 30 is

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Q7. Find the equation of the line passing through $(-3, 5)$ and perpendicular to the line through the points $(2, 5)$ and $(-3, 6)$.

Q8. Find the equation of the line passing through $(1,2)$ and $(3,-4)$.

Long Question Answers

Q1. The side of an equilateral triangle is increasing at the rate of 2 cm/s. The rate at which area increases when the side is 20 is

Q2. Solve the following system of equations

$$a) \quad 3X + 4Y = 5$$

$$2X - 2Y = 1$$

b) $3X + 5Y = 7$

$$4X + 6Y = 9$$

Q3. The tangent line to the curve $y = x^2 - 5x + 5$, which is parallel to the line $2y = 4x + 1$, also passes through another point. Determine the coordinates of this point.

Q4. The perpendicular from the origin to the line $y = mx + c$ meets it at the point $(-1, 2)$. Find the values of m and c .